PROBLEM CL1

Airfoil Wake: Streamline Displacement and Drag

Original statement · CL1.pdf

Control volume | Linear momentum | Wake | Drag

OFFICIAL PROBLEM STATEMENT

The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure of CL1.pdf. For the airfoil tested, the velocity measurements at a given downstream section reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \to -\infty, y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0, y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0, as well as the value of the drag on the airfoil as a function of ρ, y_0 and U. In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_∞ are negligible, and so is the effect of gravity on the air flow.

Source: CL1.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36, force of the fluid on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (everything per unit span), incompressible (ρ constant); neglect $\bar{g}$.
- Viscous stresses negligible in the wake and on the bounding streamlines; $p - p_\infty \simeq 0$ on the whole outer boundary of the control volume (given). On the airfoil surface pressure and viscous stresses are **not** neglected: they make the drag.

- Wake profile at $x = x_0$ (symmetric about $y = 0$): $v_x = \frac{U}{2}\left(1 + \frac{|y|}{y_0}\right)$ for $|y| \le y_0$, and $v_x = U$ outside.
- Data: $[\rho,\ U,\ y_0]$. Unknowns: $[y_{-\infty},\ D]$, with D the drag per unit span.

Symbol	Meaning	SI unit
U	uniform upstream velocity	m/s
y_0	half-width of the wake at $x = x_0$	m
$y_{-\infty}$	half-width of the stream tube far upstream	m
ρ	air density	kg/m^3
D	drag per unit span	N/m

Dimensional check of the unknown: $[\rho U^2 y_0] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m} = \mathrm{N/m}$, a force per unit length, as required for a planar problem.

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the airfoil and the tunnel; basis $(\bar{e}_x, \bar{e}_y, \bar{e}_z)$ with $\bar{e}_x$ along the stream. The control volume is fixed: $\bar{v}_c = 0$.
- Surfaces of Σ_c (outward normal $\bar{n}$): Σ_1, far upstream, $|y| < y_{-\infty}$, $\bar{n} = -\bar{e}_x$; Σ_2, the section $x = x_0$, $|y| < y_0$, $\bar{n} = +\bar{e}_x$; Σ_l, the two bounding streamlines ($\bar{v} \cdot \bar{n} = 0$); Σ_w, the airfoil surface ($\bar{n}$ pointing into the airfoil).
- Change of variable for the wake integrals: $s = 1 + y/y_0$, $dy = y_0\, ds$; $y \in [0, y_0]$ maps to $s \in [1, 2]$.

Force of the air on the airfoil (Notes Eq. 3.32), written on the wetted surface with the normal $\bar{n}$ of the control volume:

$$\boxed{\bar{F} = \int_{\Sigma_w} (p - p_\infty)\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

Hence the Σ_w terms of the momentum balance add up to $-\bar{F}$.

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-1)\,2y_{-\infty}=-2\rho U y_{-\infty}$
Σ_2	$2\rho\int_0^{y_0}\frac{U}{2}\left(1+\frac{y}{y_0}\right)dy=\frac{3}{2}\rho U y_0$
Σ_l	0 (streamlines, $\bar{v}\cdot\bar{n}=0$)
Σ_w	0 (impermeable wall)

$$-2\rho U y_{-\infty}+\frac{3}{2}\rho U y_0=0\implies\boxed{y_{-\infty}=\frac{3}{4}\,y_0}\qquad(1)$$

The Σ_2 entry in detail: the antiderivative of $1+y/y_0$ is $y+y^2/(2y_0)$, and by Barrow's rule between $y=0$ and $y=y_0$, $\rho U\left[y+\frac{y^2}{2y_0}\right]_0^{y_0}=\rho U\left(y_0+\frac{y_0}{2}\right)=\frac{3}{2}\rho U y_0$ (the factor 2 of the table counts both halves of the wake).

Linear momentum (Notes Eq. 3.36, with $p_a=p_\infty$)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_\infty)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_\infty)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-U)\,2y_{-\infty}\,\bar{e}_x = -2\rho U^2 y_{-\infty}\,\bar{e}_x$	0 ($p \simeq p_\infty$)	0 (uniform flow)
Σ_2	$2\rho\int_0^{y_0} v_x^2\,dy\;\bar{e}_x = \frac{7}{6}\rho U^2 y_0\,\bar{e}_x$	0 ($p \simeq p_\infty$)	0 (given)
Σ_l	0 ($\bar{v}\cdot\bar{n}=0$)	0 ($p \simeq p_\infty$)	0 (given)
Σ_w	0 **(impermeable)**	$-\int_{\Sigma_w}(p-p_\infty)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: the force of the airfoil on the air, $-\bar{F}$. The Σ_1 entry uses (1): $-2\rho U^2 y_{-\infty} = -\frac{3}{2}\rho U^2 y_0$.

$$-\frac{3}{2}\rho U^2 y_0\,\bar{e}_x + \frac{7}{6}\rho U^2 y_0\,\bar{e}_x = -\bar{F} \implies \boxed{\bar{F} = D\,\bar{e}_x, \qquad D = \frac{1}{3}\,\rho\,U^2\,y_0} \quad (2)$$

The Σ_2 momentum entry in detail, with $s = 1 + y/y_0$, $dy = y_0\,ds$ and Barrow's rule between $s=1$ and $s=2$: $2\rho\frac{U^2}{4}\int_0^{y_0}\left(1+\frac{y}{y_0}\right)^2 dy = \frac{\rho U^2 y_0}{2}\int_1^2 s^2\,ds = \frac{\rho U^2 y_0}{2}\left[\frac{s^3}{3}\right]_1^2 = \frac{\rho U^2 y_0}{2}\cdot\frac{8-1}{3} = \frac{7}{6}\rho U^2 y_0$. The sum of the flux column is $\left(-\frac{9}{6}+\frac{7}{6}\right)\rho U^2 y_0 = -\frac{1}{3}\rho U^2 y_0$.

For a wake symmetric about $y=0$ (a symmetric airfoil at zero incidence, as the symmetric profile implies) the y component of the momentum flux through Σ_2 cancels between the two halves, so $F_y = 0$: no lift.

PHASE 4

Interpretation, Limits and Dimensional Check

- The wake carries less mass flux ($\frac{3}{2}\rho U y_0$) than the undisturbed stream through the same width ($2\rho U y_0$): the bounding streamlines are pushed outwards, from $y_{-\infty} = \frac{3}{4}y_0$ to y_0. The momentum deficit left in the wake is the drag.
- Limit $y_0 \to 0$: no wake, $y_{-\infty} \to 0$ and $D \to 0$ (d'Alembert's paradox for a frictionless attached flow).
- Dimensions (SI): $[y_{-\infty}] = \mathrm{m}$; $[D] = [\rho U^2 y_0] = \mathrm{N/m}$.
- Numerical cross-check (midpoint quadrature of the defining integrals with 2×10^5 cells, $U = 1$ m/s, $\rho = 1.2$ kg/m^3, $y_0 = 0.35$ m): $y_{-\infty}/y_0 = 0.75000$ and $D = 0.140000$ N/m $= \rho U^2 y_0/3$.

- Agreement with the official solution: CL1.pdf obtains $y_{-\infty} = \frac{3}{4} y_0$ and $\bar{F}_{\text{airfoil}} = \frac{1}{3} \rho U^2 y_0 \, \bar{e}_x$.

Extension (beyond the official solution)

With the momentum thickness $\theta = \int_{-y_0}^{y_0} \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) dy = 2 y_0 \int_0^1 \frac{1-\sigma^2}{4} \, d\sigma = \frac{y_0}{3}$ ($\sigma = y/y_0$), the result reads $D = \rho U^2 \theta$, the classical momentum-deficit formula for the drag from a wake survey.